Achyut Katiyar

+1 (857) 576-7633 | achyutkatiyar21@gmail.com | linkedin.com/in/achyutkatiyar2103 | github.com/achyut21 | achyutkatiyar.com

Education

Northeastern University — January 2025 – May 2027
Master of Science in Computer Science — *Boston, Massachusetts, USA*

SRM University AP — October 2020 – May 2024
Bachelor of Technology in Computer Science and Engineering — *Andhra Pradesh, India*

Technical Skills

- **Languages**: TypeScript, JavaScript, Python, Java, C++, C, SQL, Bash, Solidity
- **Frameworks**: Angular 18, Vue 3, React, Next.js, Node.js, Express, Spring Boot, RxJS, Prisma, Tailwind CSS
- **Cloud**: AWS (S3, Lambda, Elastic Beanstalk, CloudFront, DynamoDB, API Gateway, Cognito), Terraform
- **DevOps / Testing**: Docker, Kubernetes, Nginx, GitHub Actions, CI/CD, Vitest, Mocha, Puppeteer, Linux
- **Databases / APIs**: PostgreSQL, MongoDB, Redis, REST APIs, WebSockets, DocuSign, OAuth2, OpenAI

Experience

Software Engineer Co-op — June 2026 – December 2026
Maven Realty — *Cambridge, Massachusetts, USA*

- Executed $1.5M+ in leases through Aptava's automated DocuSign pipeline across 18 document types, then added a DocuSign Connect webhook with HMAC verification so every deal shows live signer status.
- Built live video apartment tours on AWS IVS, with an AI chat assistant answering viewers from listing data, an AI summary of each recording, and a tour state machine that stays correct across multiple servers.
- Built mavenrealty.com in Vue 3, prerendered its 63 listing pages with Playwright for search and AI crawlers, cut page heads from 174 KB to 44 KB, and kept Zillow, RentLinx, and ADC links alive with CloudFront 301s.
- Cleared intermittent production 504s on lease PDF generation by moving an S3 existence check to HeadObject, after unread GetObject bodies pinned one socket per draft against a 50-connection pool.
- Upgraded Aptava 9 Angular major versions to 18.2 and Node 16 to 24, then rewrote every broken Material component.

Teaching Assistant – CS2000 — September 2025 – April 2026
Northeastern University — *Boston, Massachusetts, USA*

- Selected to TA CS2000 (Logic and Computation) across two consecutive semesters for a 500+ student introductory course.
- Held 4 hours of weekly office hours, graded 500+ assignments and exams, and wrote model-solution writeups on recursion, higher-order functions, and tree abstractions in Python and Pyret.

Projects

MIT Bitcoin Expo 2026 Website | *Next.js, TypeScript, Tailwind CSS, React Three Fiber, shadcn/ui*

- Drew 100k+ visitors to the 500-attendee conference site with JSON-LD structured data and per-speaker landing pages.
- Cut the initial bundle by lazy-loading the React Three Fiber star field behind SSR-safe hydration boundaries.

PayPhone | *Next.js, TypeScript, AWS, Terraform, Coinbase CDP, x402, Permit2, Anthropic API*

- Built an AI agent that bills video calls by the second through Coinbase CDP x402, at $0.01 per active second against one Permit2 witness capped at $5. 6 settlements ran on-chain, 2 of them on Base mainnet.
- Hand-rolled the x402 verify-to-settle handshake instead of CDP's `paymentMiddleware` so settlement fires on the hangup webhook, and stood up Cognito, DynamoDB, and scoped IAM in one Terraform apply.

SkillBridge AI | *Next.js, TypeScript, PostgreSQL, Prisma, Redis, OpenAI API, ElevenLabs, Docker*

- Shipped an interview prep voice coaching PWA on GPT-4 and ElevenLabs, WCAG 2.1 AA at a 95+ Lighthouse score.
- Cached GPT-4 completions in Redis to cut voice-chat latency, with Prisma Accelerate for Postgres connection pooling.

Distributed Kanban Board | *C++11, TCP, Vector Clocks, Primary-Backup Replication, Express, Socket.io, React*

- Engineered a real-time collaborative Kanban in C++ with primary-backup replication. A hot standby promotes on a 5-second heartbeat and a full state transfer restores the rejoining master.
- Resolved concurrent edits with vector clocks and last-write-wins over a custom TCP binary protocol, fronted by an Express and Socket.io gateway. p90 held under 200ms and p99 under 500ms at 50 concurrent tasks.

Maple Court Agent | *Python, MCP, vLLM, Qwen3.6, NVIDIA NemoClaw, Telegram Bot API*

- Built a Telegram property agent with 16 MCP tools and its rules enforced in code, running fully on one Dell GB10.
- Raised a 30-ticket eval from 26/30 to 28/30 and passed every security test, prompt injection included.

Leadership and Achievements

- **Hackathon Judge & Co-Chair — MIT Bitcoin Expo 2026**: Judged 47 teams alongside VCs, angel investors, and senior engineers at a 36-hour event with 150 participants and a $20K prize pool, and designed the challenge tracks.
- **Hackathon Prizes ($3,200+ across 6 wins)**: $1,500 Hacker House Goa 2024 (Nillion + Aptos tracks) | $600 Move it with Aptos | $500 Optimum Hacknet @ MIT | $500 HackSRM 5.0 | $100 Appathon 2022.